

Who Shapes the Stance? A Stases-Based Analysis of Writer-LLM Negotiation with CoAuthor

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Abstract—Large language models are pervasive in everyday writing tools, yet in most instances, the rhetorical elements of the writing are not considered. Using data from CoAuthor, an LLM writing assistant and dataset, we qualitatively coded 20 argumentative writing sessions across three dimensions: when the LLM was prompted, the rhetorical content of its suggestions, and the outcomes of those suggestions. We find a failure of negotiation where the LLMs often default to generating facts and opinions 63% of the time, which writers often reject. More critically, 33% of accepted suggestions are ultimately changed, revealing that the initial acceptance masks a deeper misalignment. Through multiple correspondence analysis and multinomial logistic regression, we find that *who* the writer is explains significantly more variance in a suggestion’s outcome rather than *what* the LLM suggests or *where* it is prompted. The solution is not to solely focus on generating better suggestions with higher acceptance rates, but build systems that learn and adapt to each writer’s unique style and rhetorical trajectory. When the LLM ignores the author’s characteristics and focuses instead on opinions, it risks displacing the writer’s own voice, underscoring the individuality each writer brings to the session. From this, we argue that LLM-in-the-loop writing tools should adopt an iterative, writer-aware design inspired by the DevOps cycle of plan, do, check, and act, to continuously align suggestions to the writer’s personal style, rhetorical trajectory, and context.

Index Terms—human-AI collaboration, writer autonomy, LLMs, rhetorical analysis, qualitative coding

I. INTRODUCTION

Argumentative writing is increasingly mediated by large language models (LLMs) as writing assistants. Most of these tools are prompt-based chatbots, which require writers to provide explicit instructions to modify the generated output. Most research on such tools uses suggestion acceptance as the primary success metric for tool efficacy [1], [2], [3]. However, writing is not just about the individual ideas or sentences; rather, what distinguishes good writing, is *how* the ideas are structured and rhetorically organized [4], [5].

Argumentative writing presents a set of ideas where claims are supported by factual evidence and personal experiences. Writers must choose what position they want to argue, how to frame their position, and how to ultimately support it. In rhetoric, writers consistently deliberate and move through five types of rhetorical choices called *stases* [6]. Stasis theory allows the writer to focus on questions of fact, definition, evaluation, cause, and propositions to effectively communicate their position [6], [7]. As this is grounded in human response mechanisms, this theory has recently been applied to other

contexts surrounding team decision making and collaboration [8], [9], so we further propose this type of investigation can be used to enhance an LLM’s suggestions.

To understand a writer’s intent, most LLM writing tools require explicit prompts, which interrupts the very flow these tools are meant to support [10]. CoAuthor takes a different approach where the writer simply presses tab to receive suggestions without providing additional information [1]. This creates a scenario of implicit negotiation; first, the LLM must infer the rhetorical intent with limited additional context, and, second, whether the writer deems these suggestions as acceptable. Understanding this negotiation is important since it reveals how suggestions can still be effective when the model is operating *blind*, or when it lacks the explicit information about the writer’s goal or style. Rather than the tool acting as a substitution for the writer, it should be used to enhance the rhetorical direction and flow of the writer’s argument.

To examine this and its failures, our work uses CoAuthor as a case study to analyze 20 argumentative writing sessions [1] across three dimensions: when the LLM was prompted, the rhetorical content of each suggestion using stasis theory [6], and the suggestion’s outcome (i.e., accepted, rejected, edited, or reversed). This analysis culminates in answering:

RQ1. During argumentative-based writing, how do authors implicitly negotiate with CoAuthor’s suggestions?

Our main finding is that individual behavioral patterns and rhetorical style explain significantly more variance in suggestion outcomes than content or where the LLM is prompted. *This has implications for designing systems that adapt to individual writers and their specific rhetorical direction.*

II. RELATED WORKS

CoAuthor, the analysis tool at the center of this research, collected over 1400 sessions from 60 different authors interacting with an LLM’s suggestions [1]. It follows best practices to increase acceptance by suggesting multiple options [11]. When the writer presses tab, CoAuthor sends the entire writing session – not including previous suggestions – to the LLM five times, and displays the first sentence of each response; unacceptable responses (e.g., empty, duplicate, vulgar) are omitted, so the writer might be shown less. After prompting CoAuthor, the writer can select at most one of the suggestions. While this work measures the average number of queries (11.8), percentage of human written text (73%), and

the overall acceptance rate of the suggestions (72%), our work focuses on the individual suggestions and their outcome.

Prior work in the writer-AI collaboration space focuses on how to enhance the writing experience. Using these tools for planning [12] and ideation [13] have been reported to improve the human experience. Other work created a cohesive writer-AI tooling design space highlighting many future areas of research [14]. Tools such as WordCraft [2], Script&Shift [15], and VISAR [3] have been designed to facilitate this collaboration process in different mediums. Unlike CoAuthor, these tools require more complex interactions and decisions from the writer, and our work aims to maintain its simplicity.

Prior work has also focused on the negative effects of LLM co-writing tools. This work concludes that authors using LLMs can be highly restrictive [16], [17] and generate less unique responses [18], [17], [13]. While LLMs have been seen to be perceived as beneficial, the actual written artifacts were not improved [19]. Even if suggestions are not accepted, maintaining a writer’s identity outside of LLM-generated text remains a prevalent challenge, and this can affect the writer’s perceived ownership [20], [21]. Our work proposes how these tools can increase the argumentative writing quality by looking at stasis theory [6] without lessening their ownership.

III. METHODOLOGY

Our work aims to identify *how* users implicitly negotiate with LLM suggestions. § III-A describes how we used the CoAuthor dataset [1]. For data analysis, we followed a mixed-method approach starting with thematic coding (§ III-B and § III-C) to prepare for Multiple Correspondence Analysis (MCA) and Multinomial Logistic Regression (MLR) (§ III-D).

A. The Dataset

The CoAuthor dataset includes keystroke-based logs of writers’ interactions with LLMs [1]. Although prior work suggests that the prompt has a small effect on collaboration [1], we selected 20 sessions from a single prompt to minimize any remaining variability. Each session was visualized as a *transcript* broken into sections of adding or removing characters, and, the focus of this paper, LLM suggestions, which can be seen in Figure 1. At the end of each suggestion section, the entire updated response, at that given time, was included. For the 20 writing sessions in this study, there were 174 instances where users prompted the LLM and 734 potential suggestions.

B. Qualitative Coding

For the suggestion blocks, we used an iterative and reflexive open-coding process, so the codes emerged from the data [22]. Through multiple rounds of refinement, we identified codes that relate to the timing, content, and outcome of each suggestion (i.e., three codes per suggestion). After external feedback from a technical communications researcher, we pivoted to a set of grounded rhetorical-based codes called stases [6], [23], which describes five main characteristics of argumentative writing. This framing allows for a reliable way

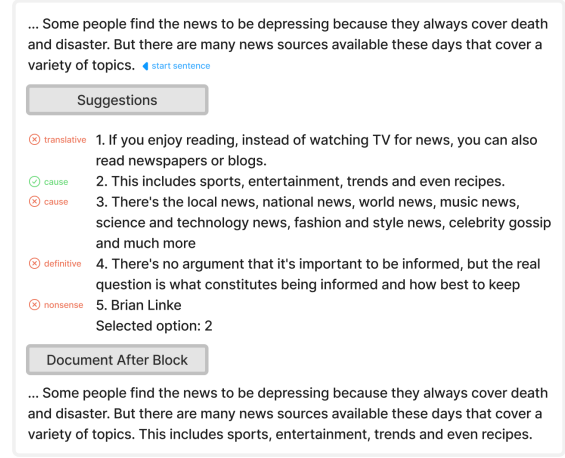


Fig. 1. Labeled Transcript Suggestion Block

to label the suggestions and has previously been used to parse argumentative essays [24], [25].

After the initial codebook was developed, two of the transcripts were coded by a second person to test its reliability resulting in 76% agreement and a Cohen’s Kappa of 0.74 [26]. The codebook was then collaboratively refined and enhanced.

C. The Codebook

This work identifies the temporal and rhetorical elements of LLM suggestions through *when* the LLM was prompted, *what* the LLM suggested, and *how* the suggestion evolved in the writing session. These dimensions are referred to as the *timing*, *content*, and *outcome* of a suggestion, respectively. This supplies our framing for the implicit negotiation between the author and LLM where timing identifies when the negotiation starts, the content of what the negotiation is over, and the outcome reveals if the negotiation succeeded or failed.

1) *Timing*: This dimension highlights *when* the user prompts the LLM for suggestions (i.e., **START SENTENCE** in Figure 1). This code represents whether prompting occurred at the **START** of a new sentence or to continue a current one so that the user stops in the **MIDDLE** of a sentence to prompt the LLM. In some instances, the user prompts the LLM, but does not select one of the suggestions, continues editing the document, and might select one of the suggestions later. These additional *skipped* instances were not included in our analysis.

2) *Content*: In classic rhetoric, stasis theory identifies five different elements in an argument and critical discourse by using guiding questions [6]. To classify the suggestions’ content, we labeled each based on where the writing should continue rhetorically (i.e., **red** or **green** text before each suggestion in Figure 1). Since LLMs are known to hallucinate [27], we also added the **NONSENSE** code to label incoherent or irrelevant instances. The **CONJECTURAL** suggestions describe the reality by introducing facts, claims, examples, or evidence (e.g., Is it? Did it happen? [8]). If a suggestion judges reality by expressing a specific stance through its importance or value, it was labeled as **QUALITY** (e.g., Of what significance

TABLE I
CO-OCCURRENCES OF CONTENT WITH OUTCOME AND TIMING. %: ROW PERCENTAGES

Content	Reject		Accept		Edit		Reverse		Total
	Start	Middle	Start	Middle	Start	Middle	Start	Middle	
Conjectural	98 (41%)	94 (40%)	18 (8%)	7 (3%)	6 (3%)	12 (5%)	2 (1%)	0 (0%)	237 (100%)
Quality	121 (53%)	64 (28%)	23 (10%)	8 (4%)	3 (1%)	5 (2%)	3 (1%)	0 (0%)	227 (100%)
Cause	43 (40%)	45 (42%)	4 (4%)	8 (7%)	5 (5%)	1 (1%)	1 (1%)	0 (0%)	107 (100%)
Translative	60 (59%)	19 (19%)	15 (15%)	3 (3%)	1 (1%)	3 (3%)	0 (0%)	0 (0%)	101 (100%)
Nonsense	32 (71%)	12 (27%)	1 (2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	45 (100%)
Definitive	5 (29%)	6 (35%)	2 (12%)	1 (6%)	1 (6%)	1 (6%)	1 (6%)	0 (0%)	17 (100%)
Total	359 (49%)	240 (33%)	63 (9%)	27 (4%)	16 (2%)	22 (3%)	7 (1%)	0 (0%)	734 (100%)

is it? [8]). The stasis element of CAUSE describes instances where it explains the reasons, consequences, or as the name suggests, the causal relationships of an argument (e.g., What caused it? [8]). TRANSLATIVE suggestions propose action through solutions or conclusions (e.g., What action (if any) is appropriate in the given case? [8]). Lastly, suggestions that explicitly define or clarify an already stated idea or topic are DEFINITIVE (e.g., What is it? [8]).

3) *Outcome*: This last dimension highlights the evolution of each suggestion, which was determined by if they were selected, changed or ultimately removed from the final state of the writing session (i.e., ✖ or ✔ in Figure 1). Suggestions that were ignored or not selected were classified as a REJECT. If an LLM suggestion was selected, three outcomes emerged. First, the accepted suggestion might remain unchanged for the rest of the writing session which we label as ACCEPT. Second, if the author later made updates to the suggestion, it was labeled as EDIT, indicating partial alignment. Third, an accepted suggestion that was then entirely removed was labeled as REVERSE, which shows misalignment with the writer’s rhetorical intent.

D. Multi-dimensional Analysis

The three dimensions – timing, content, and outcome – of each suggestion are inter-related. We used co-occurrences, multiple correspondence analysis (MCA), and multinomial logistic regression (MLR) to verify this. Our p-value significance threshold is $\alpha = 0.05$. The MCA comparison of timing and content was supported with a chi-square analysis. Given our imbalanced data based on the outcome dimension, the chi-square analysis of content and outcome used a binary analysis where the four outcome codes were grouped based on their final state. For example, all suggestions that were ultimately kept (i.e., ACCEPT and EDIT) were compared to the ones who were not (i.e., REVERSE and REJECT). However, for MLR, the outcome had three potential outputs revolving a stable ACCEPT, unstable accept (i.e., EDIT or REVERSE), and REJECT. This decision allowed us to keep a level of uncertainty when it comes to the LLM suggestions since an accepted suggestion that never changes and one that is ultimately reversed should be weighted differently.

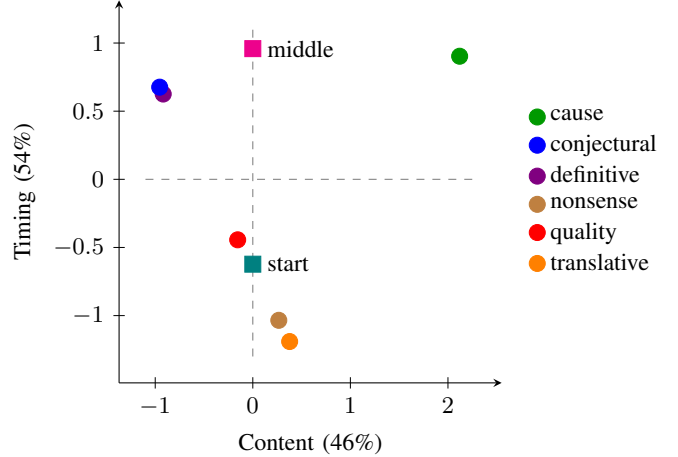


Fig. 2. MCA Biplot of Content $\times$ Timing

IV. RESULTS: THE INTERPLAY OF DIMENSIONS

While § III-C introduces the dimensions individually, their importance lies in their interactions (Table I). We examine how timing and content correlate (§ IV-A), investigate what predicts suggestion outcomes (§ IV-B), and whether the LLM’s temperature affects content generation or outcomes (§ IV-C).

A. Where and What: LLM Generation Patterns

We used MCA to analyze the relationship between timing and content [28] Chi-Square analysis tested the independence of these relationships. This pairing shows a significant relationship ($p < 0.001$) with a small to moderate effect size (Cramer’s $V = 0.194$). This confirms that *where* the LLM is prompted influences what it generates.

Figure 2 shows two clusters based on the timing (y-axis) and the content (x-axis). Both dimensions are similarly important, explaining 54% and 46% of the variance, respectively. The axis ranges reflect the spread of the MCA factor scores which are centered at zero and spread depending on how strongly the stasis categories relate to each dimension.

The TRANSLATIVE, QUALITY, and NONSENSE codes are clustered closer to the START of the sentence. The second cluster, for when the LLM is prompted in the MIDDLE of a sentence, skew closer to DEFINITIVE, CONJECTURAL, and

TABLE II
MULTINOMIAL LOGISTIC REGRESSION MODEL COMPARISON.
** : SIGNIFICANT COMPARED TO M1

Model	Predictors	Log-Lik	McF. R ²
M0	Null (intercept only)	−436	.00
M1	Content + Timing	−424	.03
M2	Content * Timing	−417	.04
M3**	Content + Timing + Writer	−396	.09

CAUSE, but is spread out along the x-axis. CAUSE highlights its uniqueness as it is the only stasis that uses relational-based reasoning between two different elements compared to the more explanatory power of the other stasis elements. The y-axis provides additional insight into rhetorical organization where positive values relate more to causal or explanatory reasoning whereas the negative values are more definitive evaluative, or claims-based.

B. Who: Writer Identity as the Predictor of Outcomes

MLR shows how the content and timing dimensions predict the outcome [29]. Due to the sparsity of definitive suggestions ($n = 17$), it was combined with the conjectural stasis due to their similarity (Figure 2). Table II summarizes the various models that we tested so that the log-likelihood, a measure of how well the model fits the data, should be optimized for the least negative value [29] and the McFadden’s pseudo- R^2 references the explainability difference between the predictors where the larger numbers are better [30].

While M2 does increase the models fit ($-424 \rightarrow -417$) and explainability ($0.03 \rightarrow 0.04$), it is not significant. Common practice suggests a more complex model should not be considered if it is not significant [31]. On the other hand, M3 measures the effect of *who* the author was; for the 20 sessions, there were four authors who wrote two sessions giving us 16 unique authors. By adding this predictor, it significantly improved the models fit and increases the explainability where 9% of the variance in the outcome dimension is attributed to the author rather than the content and location dimensions.

C. CoAuthor’s Different Temperatures

A distinguishing factor of the CoAuthor dataset is its test of two temperatures, 0.2 and 0.9, for the LLM’s prompting variation. For our study, there are nine sessions with the 0.2 temperature (209 suggestions) and eleven sessions use 0.9 (525 suggestions). In this section, we test whether this affects the content and outcome dimensions.

First, we compare the stasis distribution across temperatures, (Table III). CAUSE, CONJECTURAL, DEFINITIVE, and NON-SENSE double in frequency when the temperature increases. TRANSLATIVE decreases slightly, and QUALITY is halved.

The Chi-Square analysis is significant ($p < 0.001$), and 0.3 Cramer’s V describes a strong relationship between temperature and stasis. There is no significance for outcome and temperature implying that focusing on tuning the temperature alone is insufficient to improve the author’s experience.

TABLE III
CONTENT DISTRIBUTION BY TEMPERATURE

Content	Temp 0.2	Temp 0.9
Cause	18 (9%)	89 (17%)
Conjectural	41 (20%)	196 (37%)
Definitive	2 (1%)	15 (3%)
Nonsense	6 (3%)	39 (7%)
Quality	108 (52%)	119 (23%)
Translative	34 (16%)	67 (13%)
Total	209 (100%)	525 (100%)

V. DISCUSSION & DESIGN IMPLICATIONS

This work reveals that the LLM’s blindness to the writer’s rhetorical trajectory causes failed negotiations. Writers rejected 82% of the suggestions, and 23% of the suggestion blocks resulted in no selection. Even accepted suggestions are unstable (i.e., edited or reversed) 33% of the time. Our findings show that the writer explains more variance in outcomes. Writing tools should adapt to the individual writers. To maintain the non-intrusive design benefits of CoAuthor, we propose using an iterative DevOps cycle – plan, do, check, and act (PDCA) [32] – to align suggestions with the emerging rhetorical trajectory and personal style.

A. LLM Temperature as a Design Choice

As our results show, temperature significantly changes what the LLM generates, but does not necessarily improve outcomes. However, temperature may play a supporting role in a writer-aware system by dynamically adjust the temperature to match the writer’s current needs. Based on Table III, lower temperatures would procure actionable solutions (i.e., TRANSLATIVE, QUALITY) and higher temperatures are used for exploration or ideation (i.e. CAUSE, CONJECTURAL).

B. Conjectural & Quality Suggestion Dominance

Across the 20 writing sessions, 63% of the suggestions were either labeled with CONJECTURAL or QUALITY. While argumentative writing does rely on factual claims and evaluative stances, this dominance shows that the LLM tries to lead the argument rather than supporting the writer’s. The other stases enable better arguments through justification, causal reasoning, and actionable conclusions. This shows that the LLM is not adapting to the writer’s rhetorical needs. For example, a writer establishing a scenario needs different support compared to one seeking evidence. A writer-aware design would identify where the writer is in the argument and adjust the suggestion types accordingly.

C. The Power in Keeping a Suggestion

Around 33% of the accepted suggestions were later changed or removed, highlighting that acceptance does not necessarily imply successful negotiation. While the edited suggestions are relatively split across the start (16) or middle (22) of the sentence the majority are CONJECTURAL (18/22). The writer wants to verify that the current state of their argument matches

the description of a suggestion’s reality, or vice versa. In cases that were reversed, all were at the start of a sentence, and three of the seven were *QUALITY*. This implies that while the LLM suggested a judgment about reality, it was a misinterpretation. While there are only a few instances of reversing suggestions, we believe reversals happen less in the middle of a sentence because the LLM can use what has already been written to guide the suggestions. This observation offers a design implication, that when the LLM is prompted at the beginning of the sentence, it should refer to the rhetorical elements of the previous sentences.

1) *Subsequent Effects of Regretting Decisions:* The majority of *EDITED* suggestions identify instances where the writer made small changes (i.e., fixing grammar, merging sentences, moving suggestions) which implies that the suggestions were almost there. However, many edited suggestions show that the writer wanted to continue the idea by removing the terminal punctuation. In these instances, it offers an interesting perspective where the writer uses the suggestion as a starting point rather than a conclusion.

For *REVERSED* suggestions, more often than not, it seems that the writer’s direction was not directly influenced by the suggestion. Although, one of the writers substantially reshaped the surrounding content to fit the suggestion into their narrative, but then ultimately decided to remove it.

D. Future Work

To help argumentative writers, tools must go beyond treating acceptance rates as the primary metric for determining its success. A suggestion’s success or failure is largely determined by the specific author, so future work should focus on uncovering the writer’s unique characteristics. We propose combining stasis theory [6] with PDCA [32] to personalize the writer’s experience (§ IV-B). When prompted, the LLM should:

- 1) **Plan.** Based on previous interactions (i.e., sessions or negotiations), the LLM should leverage contextual insights and our stases-based findings (e.g., query location, logical argumentative flow, and temperature-based content distributions) with an LLM-as-a-judge quality check to produce suggestions that better support the author.
- 2) **Do.** Provide the suggestions.
- 3) **Check.** Based on the author’s response, there should be considerations for *why* an option was rejected, selected, or edited. This check stage is also continuously referenced to track the evolution of suggestions.
- 4) **Act.** After verification, adjustments to the writer’s profile, stasis-dependent relative temperatures, and the LLM’s context should be updated for future reference.

Beyond the design decisions related to *how* the LLM reacts to prompting and suggestion selection, there should be a focus on *what* the author wants to convey. Similar to prior work [15], an iterative planning stage prior to writing will tailor the suggestions to fit the author’s rhetorical trajectory, and the tool can use stasis theory to help enhance the discussion.

VI. THREATS TO VALIDITY

Internal Validity The majority of this work was completed qualitatively and by the first author. This offers a level of inherent bias to the interpretations. To counteract this, the qualitative codebook has .74 level of agreement [26]. Similarly, the stases-based motivation of this paper was verified by a technical communication researcher.

External Validity. While we analyzed a small sample of the CoAuthor dataset and one prompt, their analysis concluded that different prompts had minimal impact on the LLM’s suggestions [1]. We kept the same prompt to mitigate the potential variance of specific rhetoric elements across different prompts. However, stasis theory is more about the reasoning behind the sentence or phrase rather than the actual content, so we believe these findings are transferrable [24], [25]. Similarly, the 16 authors’ results are not necessarily generalizable, but do offer insight for applications to a broader population.

The CoAuthor dataset was released in 2022, and LLMs have changed, so the transferability of our findings to newer models might differ. However, we believe our observations of PDCA can transfer to any model because it focuses more on the user experience. Similarly, based on prior work, performance is relatively similar across the top models [33].

Construct Validity. The qualitative codes are granular which offers a less fine-grained analysis of how humans and LLMs implicitly negotiate for argumentative writing. Similarly, highlighting when the LLM was prompted does not include why the human wants help from the LLM, however this is beyond the scope of this dataset. Similarly to this granularity, based on the limited set of writing sessions, while we believe that knowing the writer significantly improves the curation of suggestions, this is still largely understudied.

Conclusion Validity. The codes in the different dimensions are largely unbalanced. There was a majority of *REJECT* and minimal representation of *DEFINITIVE*. Having a more balanced dataset could affect our analysis, but this supports one of our key findings of an LLM’s preference of specific stases-based codes. Similarly, much of this paper focuses solely on a sentence-level analysis which limits overall session findings; however, § V-C1 highlights some preliminary investigation into what happens beyond a suggestion’s selection.

VII. CONCLUSION

LLMs are pervasive in writing tools, yet there is little work on how iterative context improvement can affect the rhetorical structure of argumentative writing. Rather than persuade the author towards a specific type of argument, the LLM should be used as a tool to enhance the person’s claims. This work uses CoAuthor for its simplistic design to understand the implicit negotiation between the LLM and writer. We identify fallacies based on the over-inflation of *QUALITY*- and *CONJECTURAL*-based suggestions that provide the author with a specific narrative rather than in support of one. Argumentative writing is a powerful tool, and by using the PDCA principles, LLMs can iteratively refine suggestions to be more coherent and rhetorically aligned with what the writer wants.

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